
Building a Digital Corpus of Euclid's *Elements* Title Pages

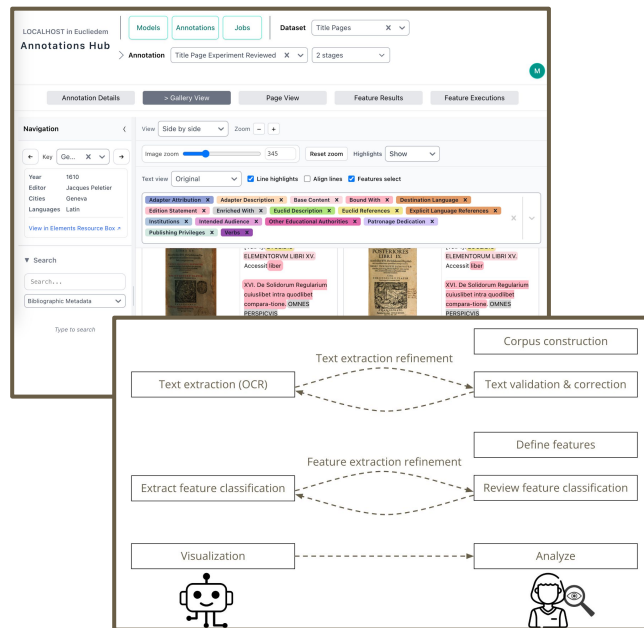
Methods and Findings

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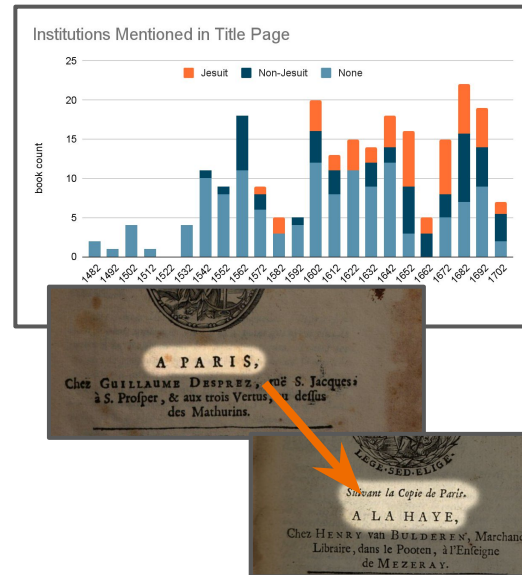
Presentation Plan

1. Motivation & Research Questions

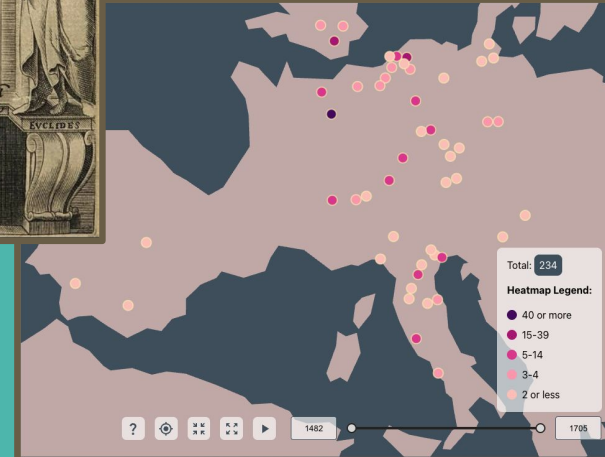
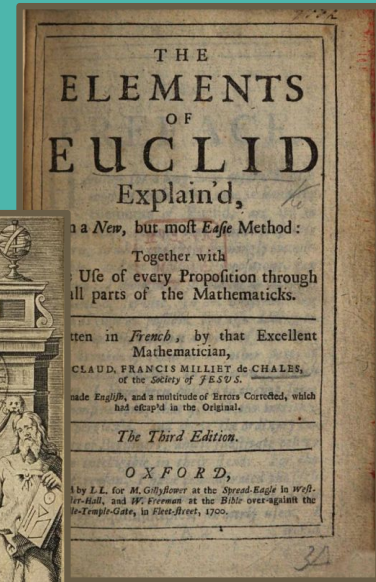
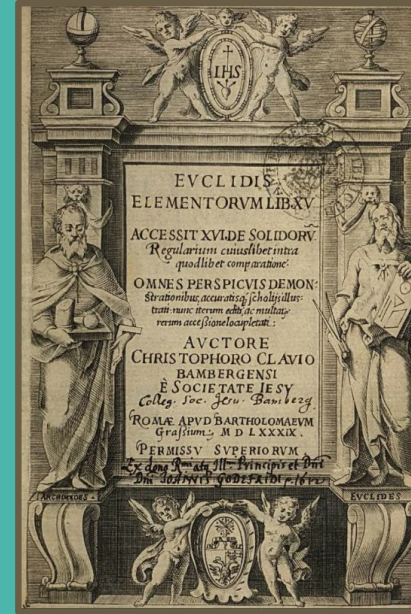


2. Approach, Methodology & Platform Walkthrough

3. Findings



Motivation & Research Questions



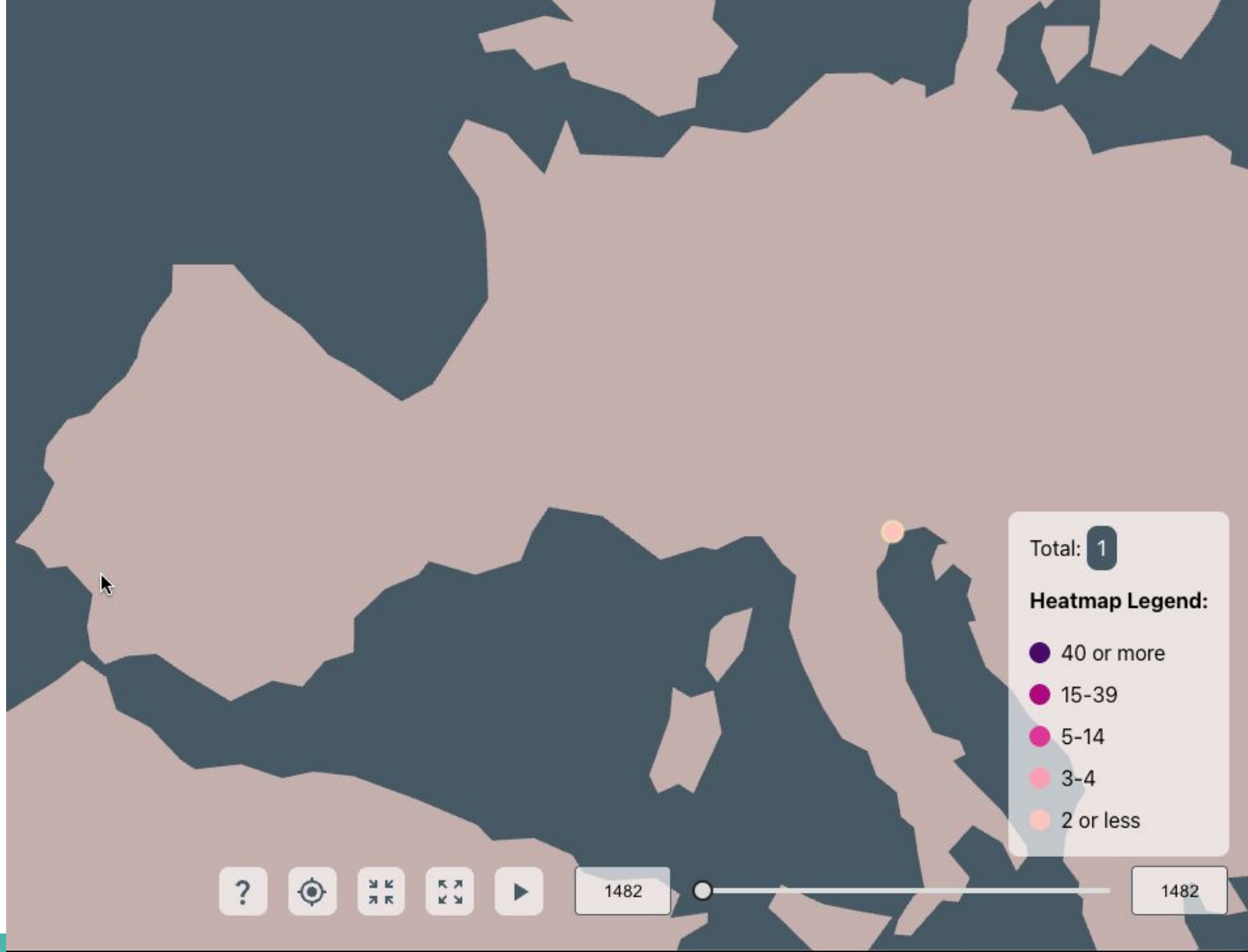
Why Euclid's *Elements* ?

- A foundational mathematical text since antiquity.
- 16th-17th centuries: Held a canonical status in early modern Europe.
- Wide circulation across multiple contexts: from highly practical applications to the intellectual milieus of learned Europe.
- Translations varied in presentation, commentary, and mathematical content.

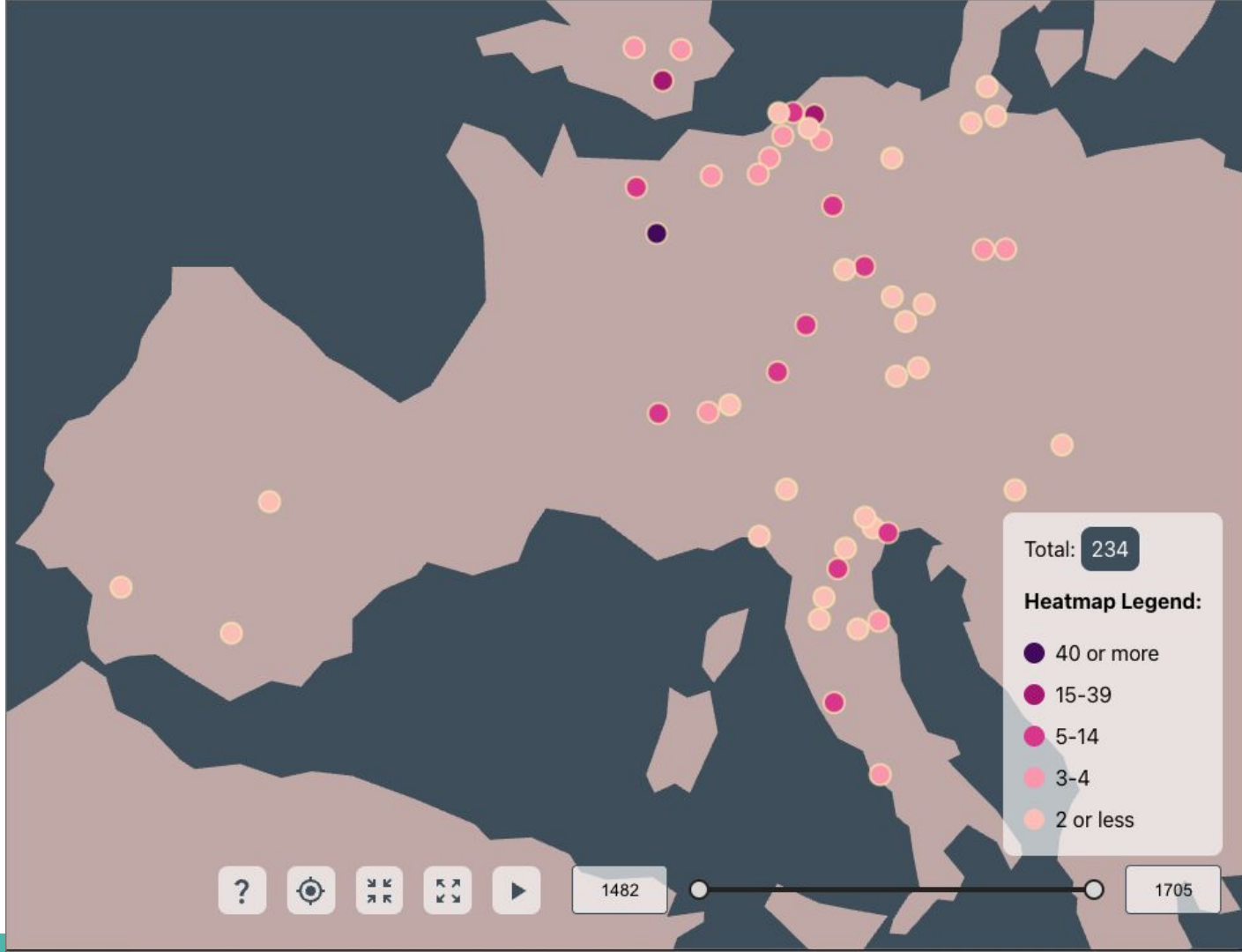
→ Offers a window into the broader mathematical landscape of the time.



A Proliferation of Editions

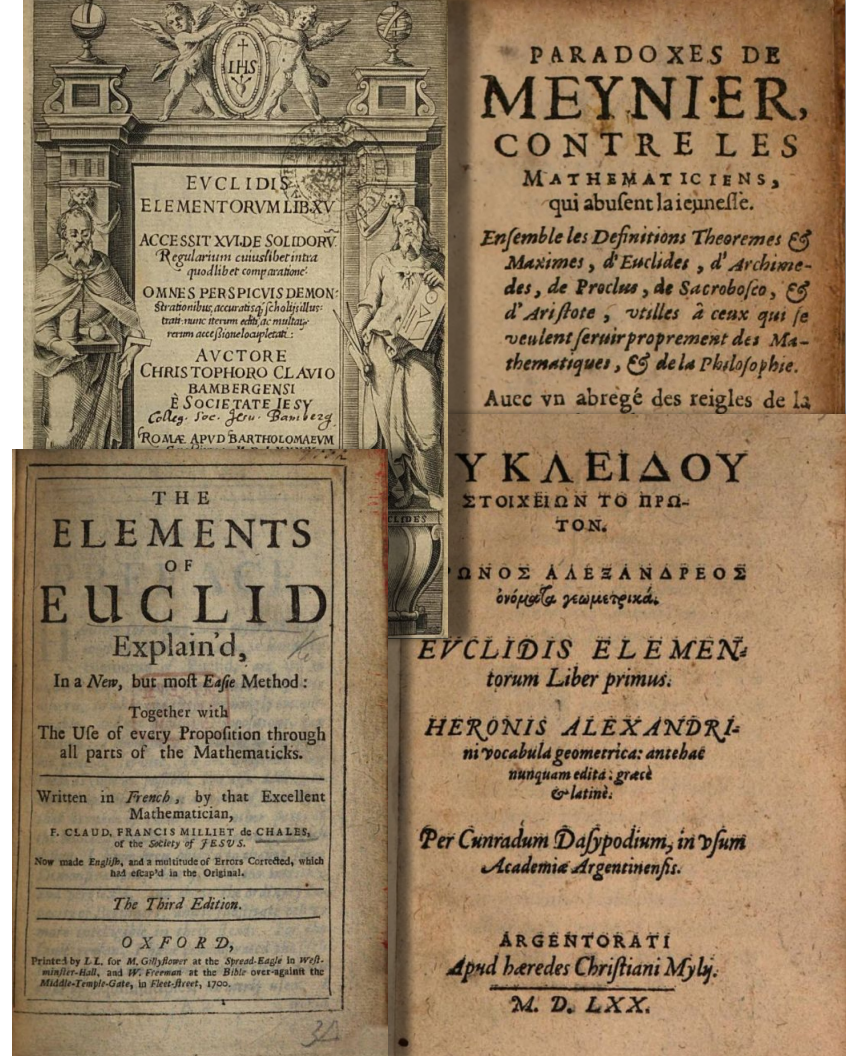


A Proliferation of Editions



Why Title Pages?

- A condensed statement of purpose: exposes multiple layers of information.
- Reflect how the book was positioned and circulated.
- Shed light on the social, intellectual and commercial forces shaping its transmission.
- Offer a panoramic overview.

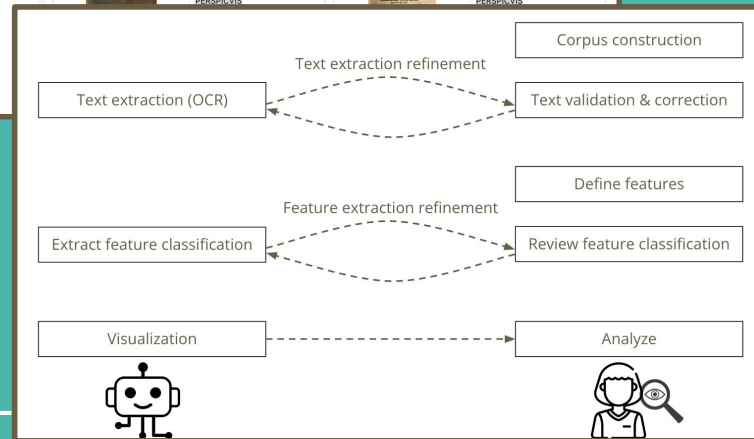


The Research Question

How the title pages of early modern printed editions of Euclid's *Elements* reflect its intellectual and social positioning?

Approach, Methodology & Platform Walkthrough

The screenshot displays the 'Annotations Hub' interface. At the top, there are tabs for 'Models', 'Annotations', 'Jobs', and 'Dataset'. The 'Dataset' tab is selected, showing 'Title Pages' with a dropdown menu. Below this, there's a section for 'Annotation' with a dropdown menu set to 'Title Page Experiment Reviewed' and a '2 stages' dropdown. The main area is divided into 'Annotation Details', 'Gallery View' (selected), 'Page View', 'Feature Results', and 'Feature Executions'. The 'Gallery View' shows a list of annotations with fields like 'Key', 'Ge...', 'Year', 'Editor', 'Cities', and 'Languages'. A search bar is visible on the left. The right side shows a detailed view of an annotation with various fields like 'Adapter Attribution', 'Adapter Description', 'Base Content', 'Bound With', 'Destination Language', 'Edition Statement', 'Enriched With', 'Euclid Description', 'Euclid References', 'Explicit Language References', 'Institutions', 'Intended Audience', 'Other Educational Authorities', 'Patronage Dedication', 'Publishing Privileges', and 'Verbs'. Below this, there are images of book pages with annotations.



Methodology

- **Corpus:** 219 editions of *Elements* published before 1700.
- **Focus:** Segmenting the text of each title page into distinct elements to identify patterns and variations.
- **Theoretical Framework:** Based on Gérard Genette's structuralist ideas about paratexts.
- **Strategy:** Building a reusable researcher-in-the-loop platform that identifies and classifies text parts and enables large scale, systematic and flexible comparison.

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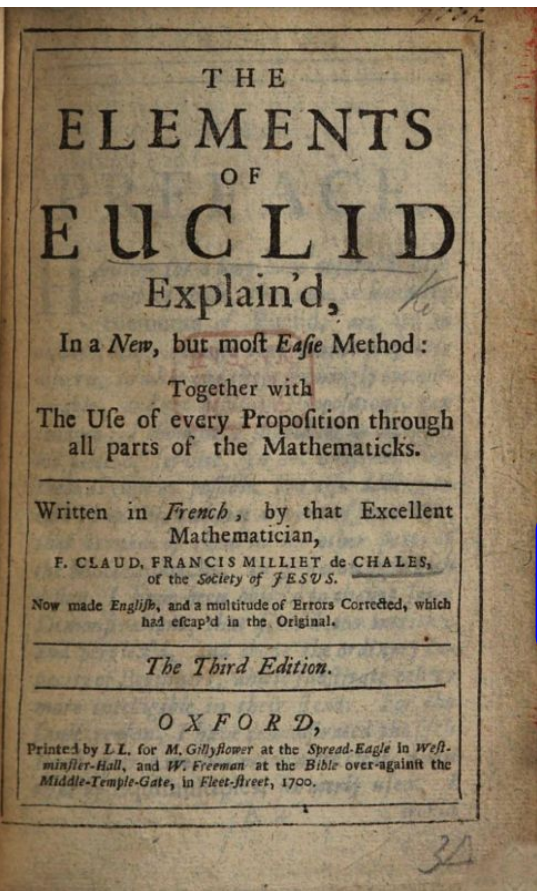
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Segmenting the Title Page's Text



THE
ELEMENTS
OF
EUCLID

Explain'd,

In a New, but most Easie Method:
Together with The Use of every Proposition through
all parts of the Mathematicks.

Written in French, by that Excellent
Mathematician,

F. CLAUD. FRANCIS MILLIET de CHALES,
of the Society of JESUS.

Now made English, and a multitude of Errors Corrected, which
had escap'd in the Original.

The Third Edition.

OXFORD,

Printed by L.L. for M. Gillyflower at the Spread-Eagle in West-
minster-Hall, and W. Freeman at the Bible over-against the
Middle-Temple-Gate, in Fleet-street, 1700.

Base Content

Elements Designation

Base Content
Description

Adapter action verbs

Languages

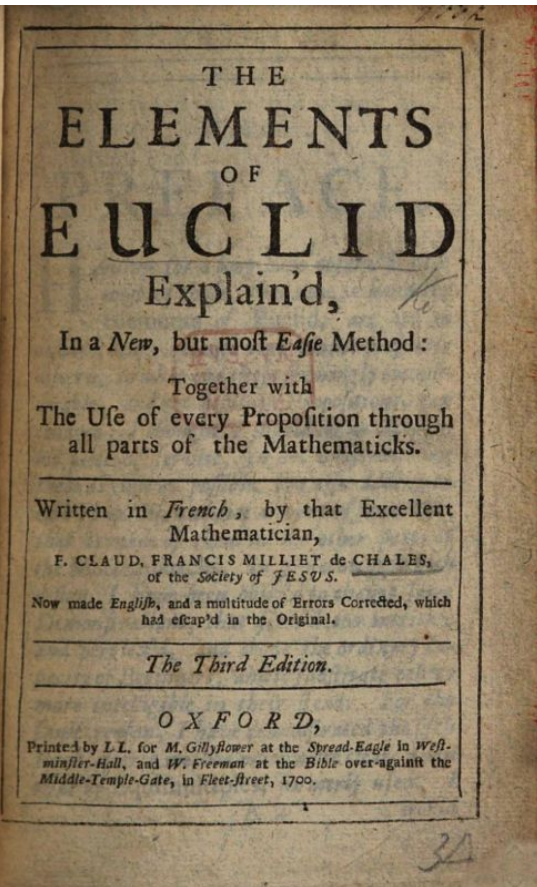
Institutions

Adapter Description

Edition Statement

...

Segmenting the Title Page's Text



THE ELEMENTS OF EUCLID Explain'd

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THE ELEMENTS OF EUCLID

Elements Designation

In a New, but most Easie Method: Together with The Use of every Proposition through all parts of the Mathematicks.

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Adapter action verbs.

French, English

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Excellent Mathematician, ..., of the Society of JESUS

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Now made English, and a multitude of Errors Corrected, which had escap'd in the Original. The Third Edition.

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Research Platform



<https://euclides.huma-num.fr/hub>

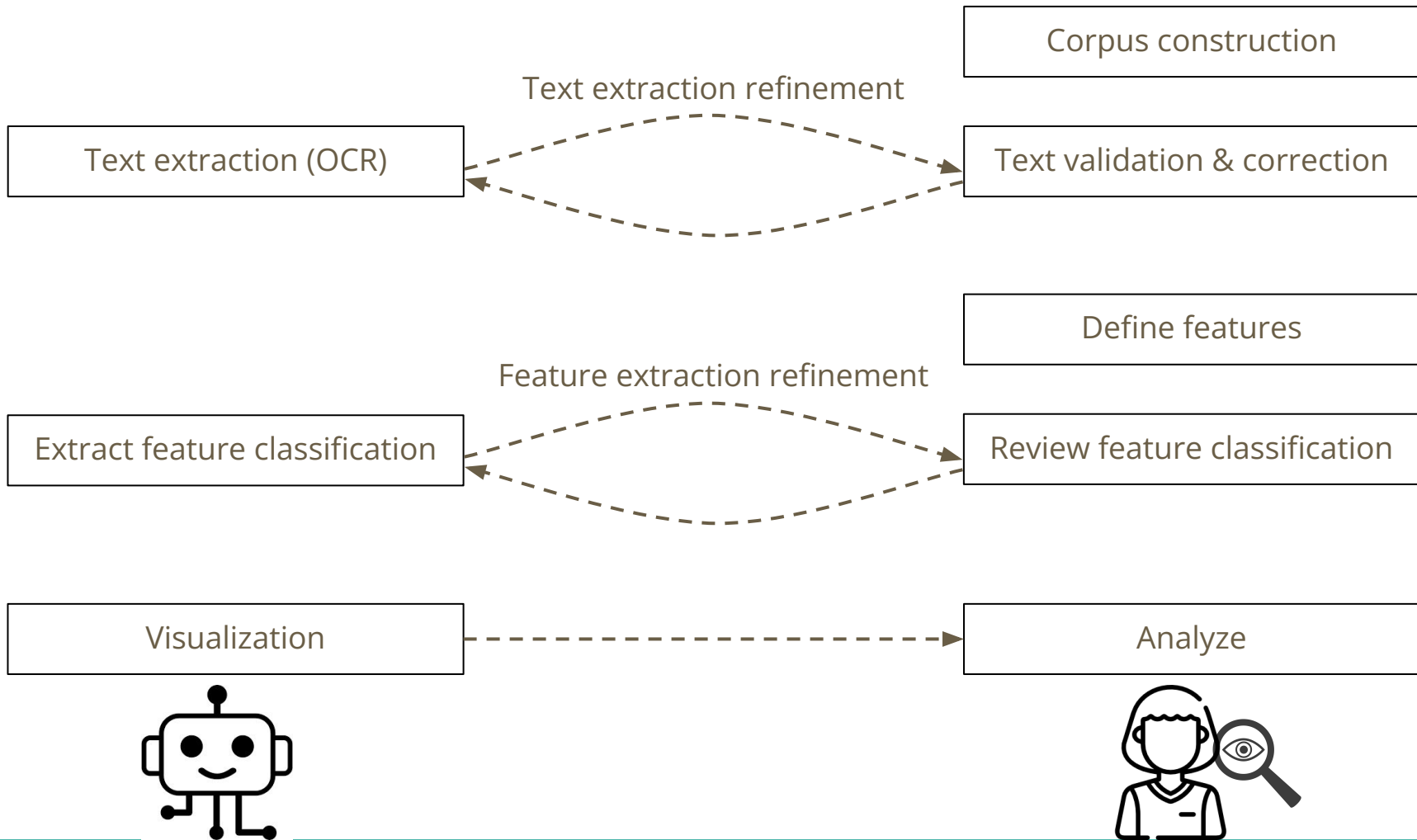
Title Page Results



<https://elements-resource-box.netlify.app/titlePages>

Scan to Follow Along!

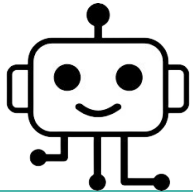
⚠ Work in progress! ⚠
Might break, disappear, or evolve.



Text extraction (OCR)

Extract feature classification

Visualization



Corpus construction

Text validation & correction

Define features

Review feature classification

Analyze



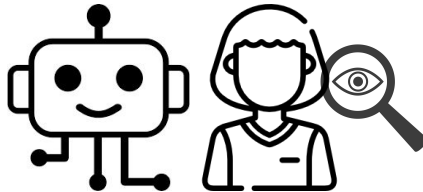
Text extraction refinement

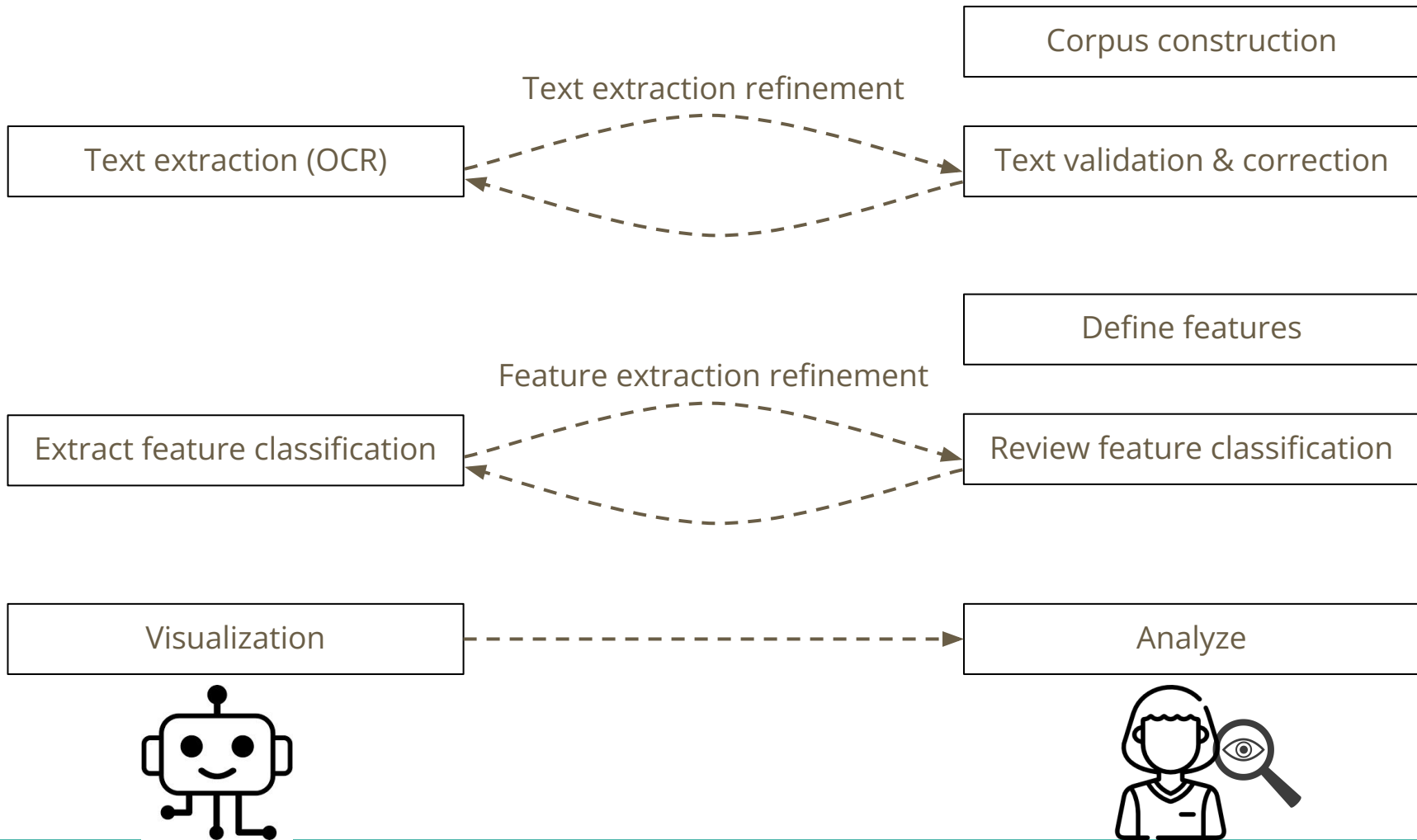


Feature extraction refinement



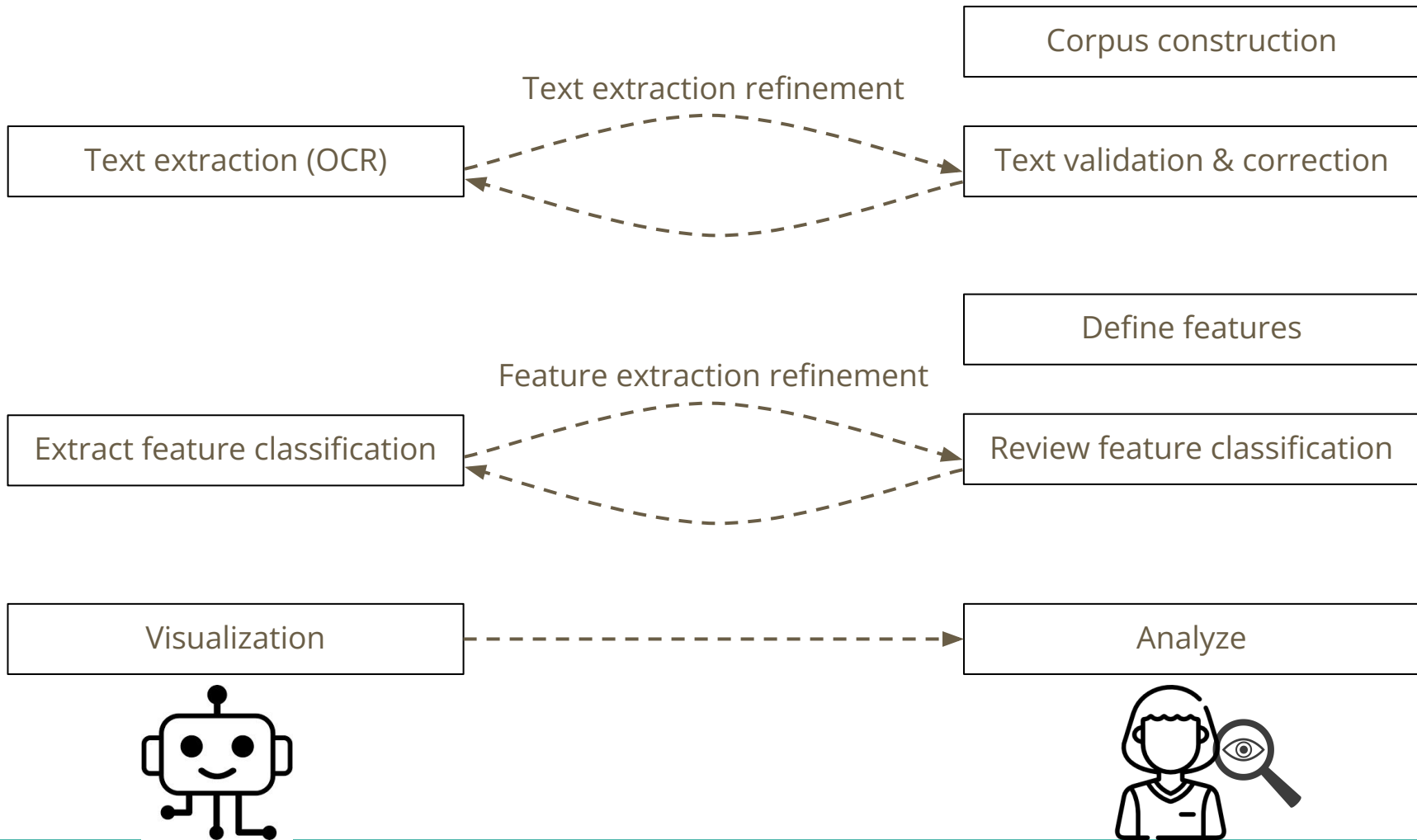
Informed Analysis

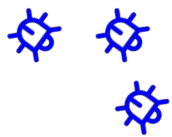




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I have a another research
Q (and corpus)!

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I should add this
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Text validation & correction

This new OCR method
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What about this feature?

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Bug in the code!

Visualization



Analyze



Research Platform



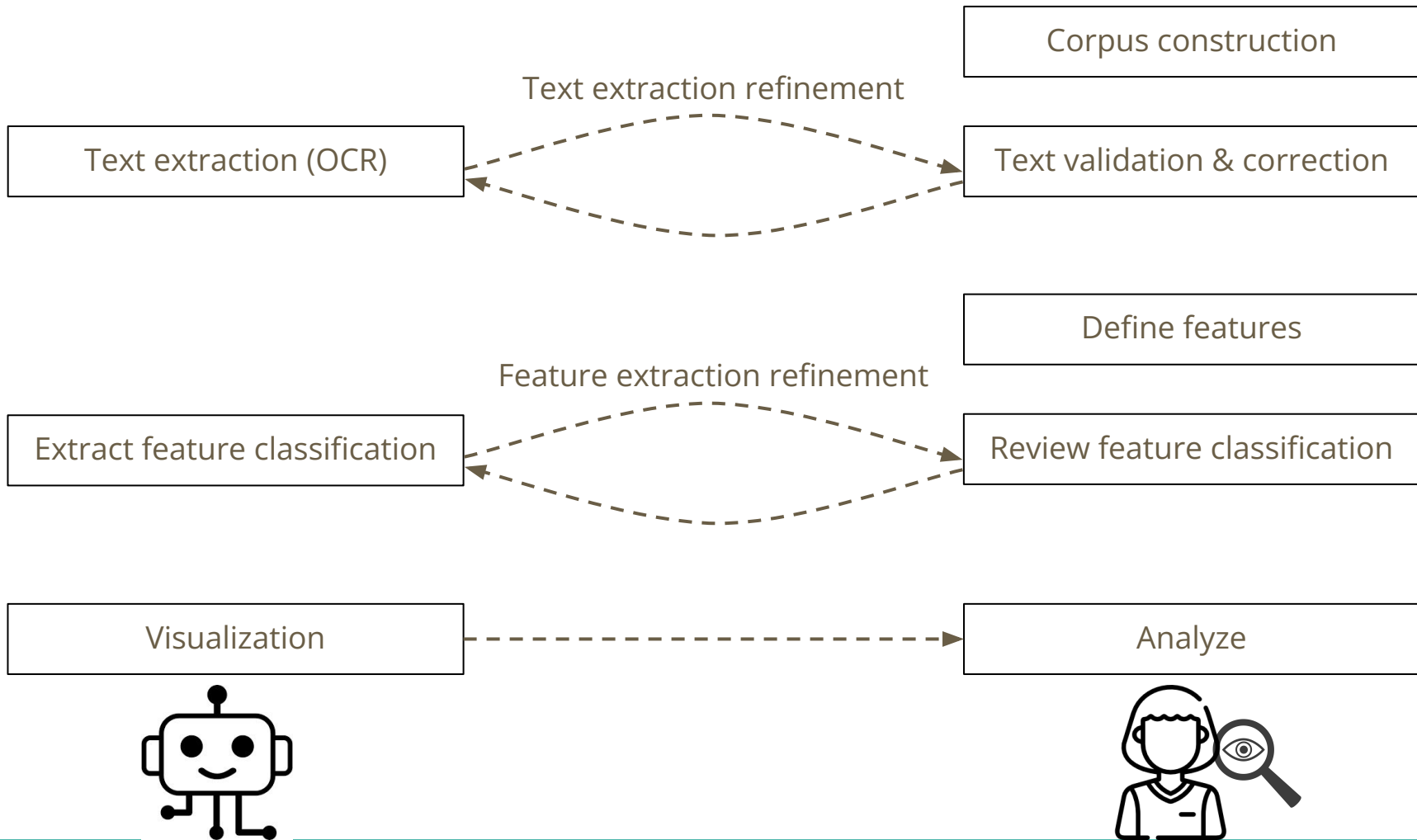
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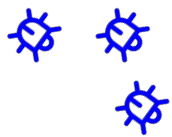
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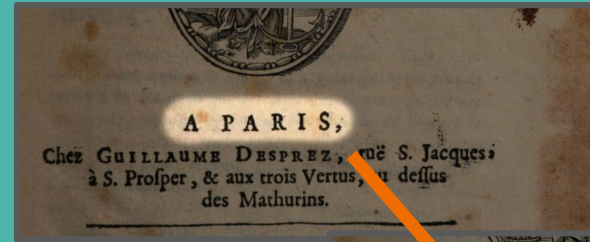
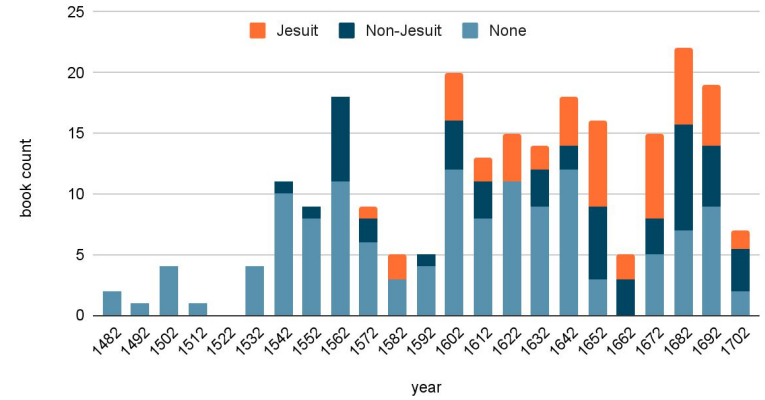


Analyze



Findings

Institutions Mentioned in Title Page

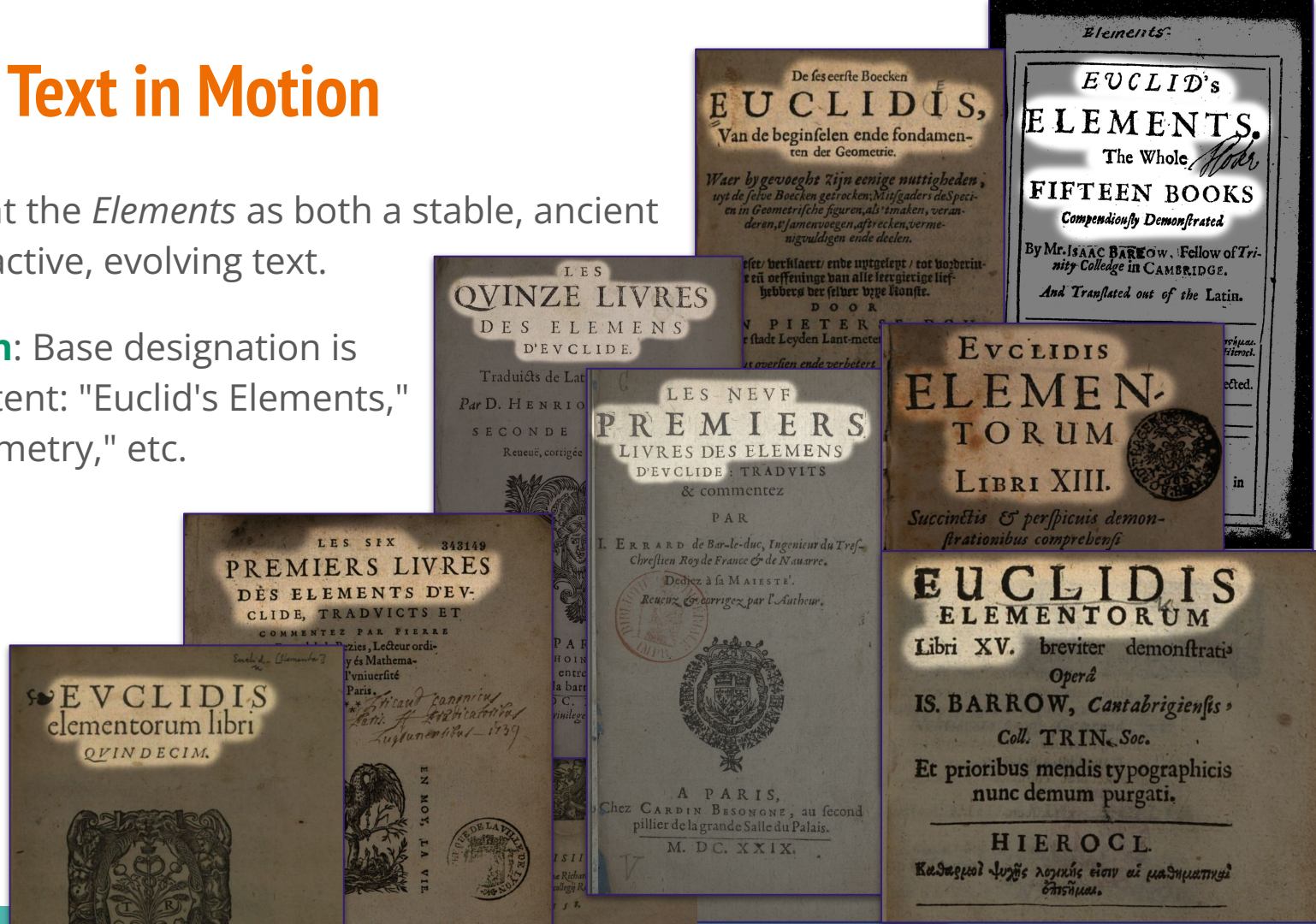


A Canonical Text in Motion

Title pages present the *Elements* as both a stable, ancient authority and an active, evolving text.

- **Stable Canon:** Base designation is highly consistent: "Euclid's Elements," "Euclid's Geometry," etc.

Title pages, 1558-1705,
published across Europe in
English, Latin, Dutch and
French

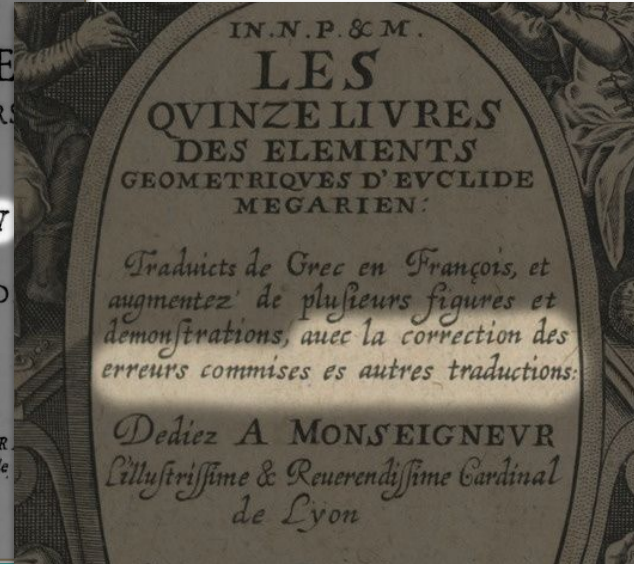
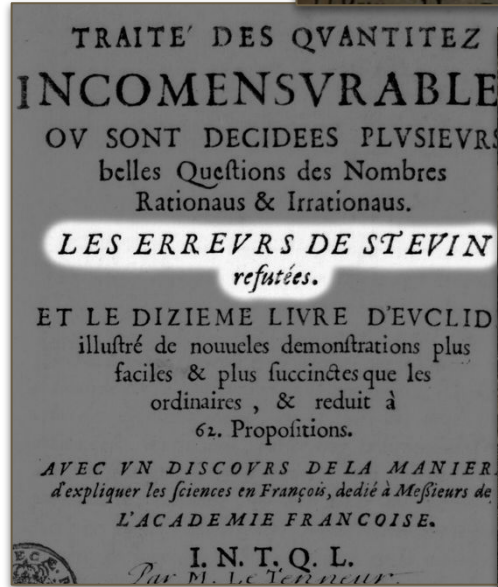
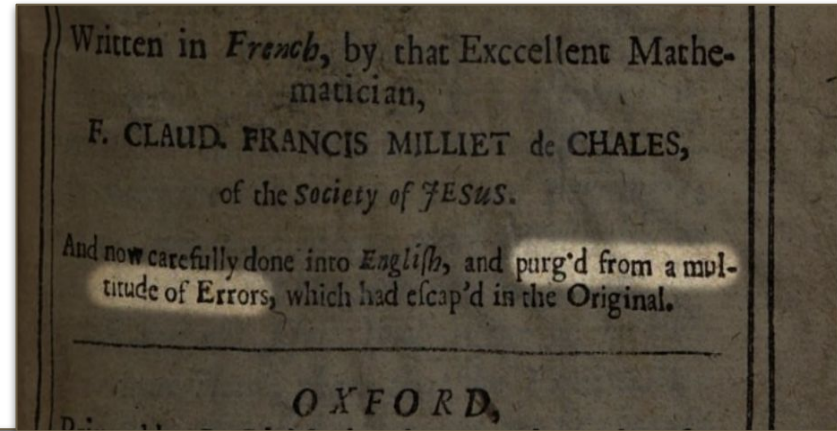


A Canonical Text in Motion

- **Text in Motion:** Title pages often reference corrections improvements and additions over previous editions and adapters.

- A single, ancient, consolidated work that retain its identity as a definitive mathematical authority.

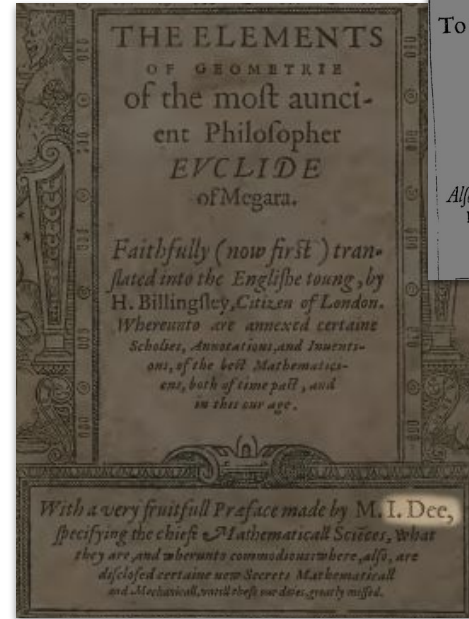
- And an active player of an evolving, contested mathematical landscape.



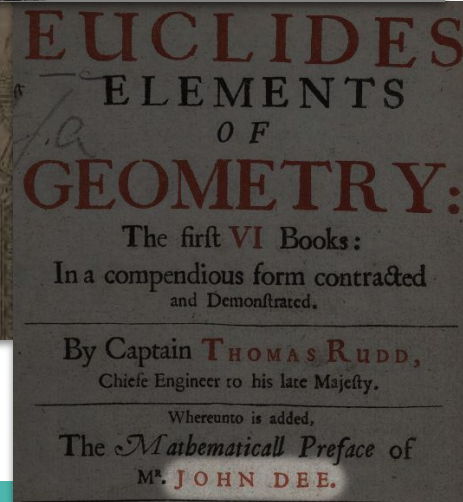
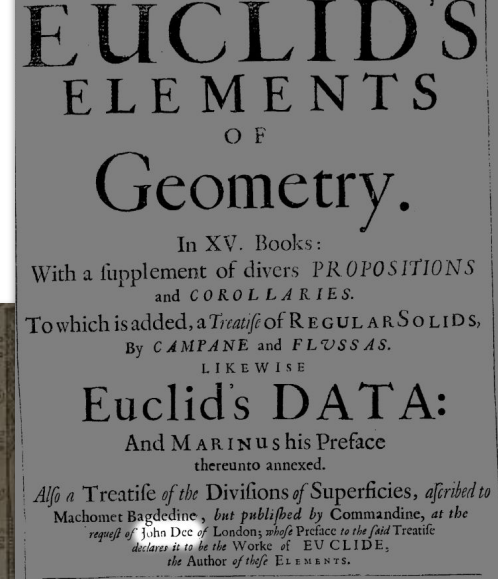
Local Roots and Cross-European Circulation

Publishing operated on two scales simultaneously:

- 70 of 97 adapters published in only one city.
- Title pages reference local intellectual lineages and institutions.

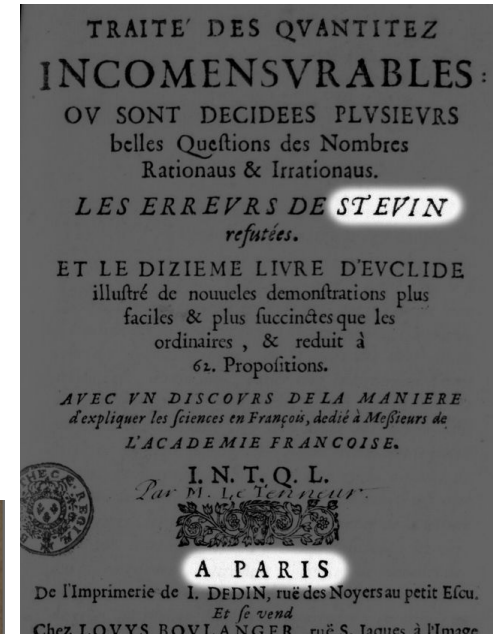
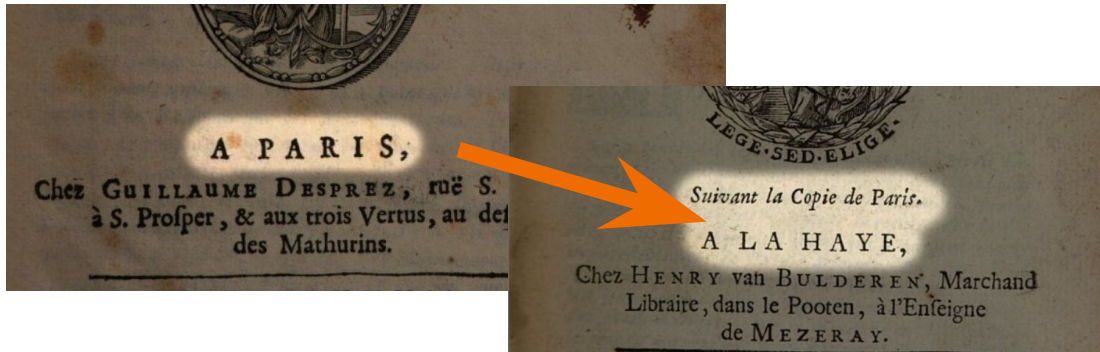


Title pages of London editions published between 1570 and 1651 that mention John Dee.



Local Roots and Cross-European Circulation

- Mathematical publishing operated across national borders.
- Imprints and references reveal active exchange.
- The Elements act as a vehicle of continental scholarly dialogue.

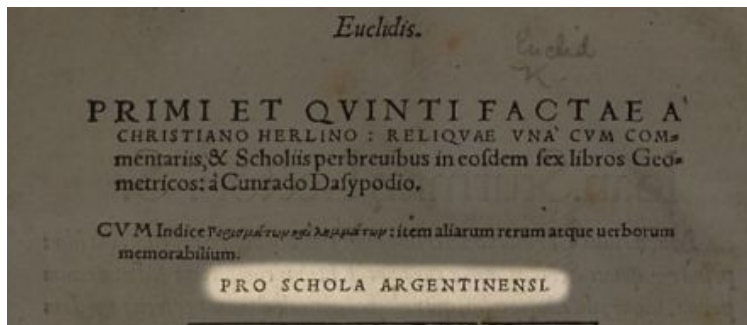


Institutional Affiliations and Mentions

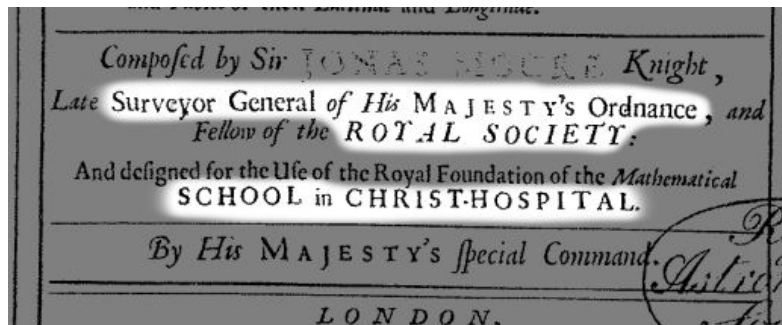
Title pages place editions within specific institutional contexts, which establishes the adapter's authority.

These contexts were diverse:

1. Universities, colleges and schools



2. Civic, commercial and military institutions



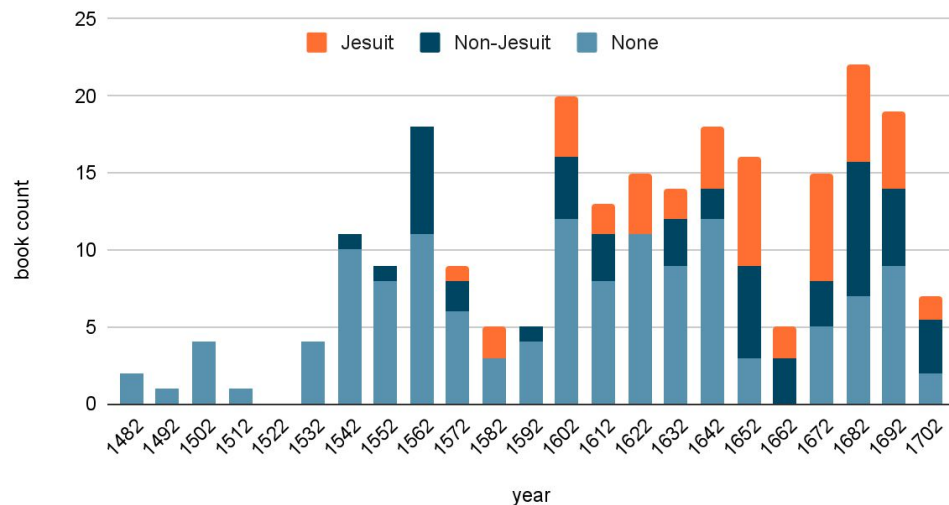
3. The Jesuit Order



The Jesuit Order: A Pan-European Institution

- A prominent pan-European affiliation.
- Often used as a sole credential of the adapter.
- Vernacular title pages mention Jesuit identity far less often, revealing sensitivities tied to local religious and political contexts.

Institutions Mentioned in Title Page

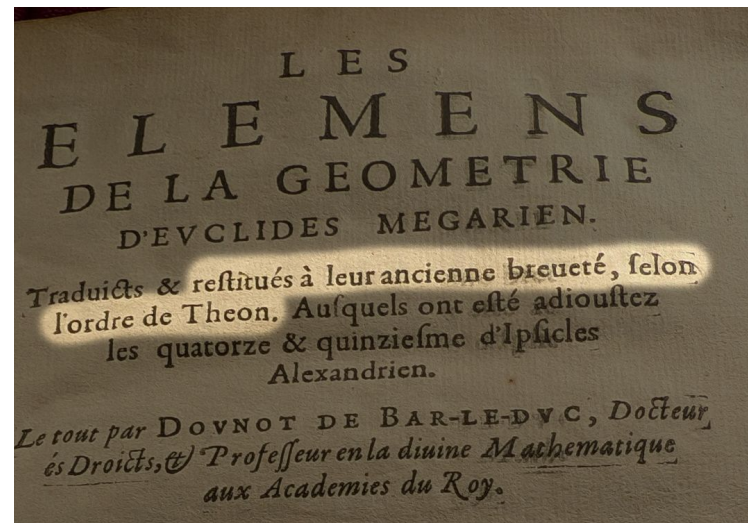


Between Restoration and Adaptation

Title pages reveal a tension between two competing values:

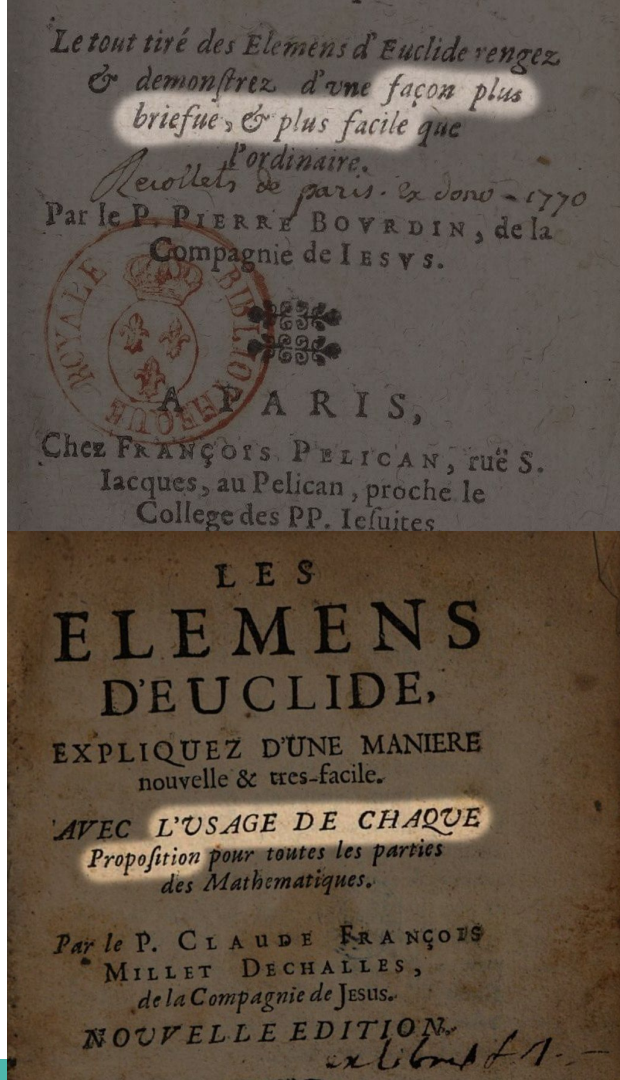
Restoration:

- A focus on philological purity and recovering the "true" ancient text.
- Noting translations "from Greek."
- The adapter is presented as a restorer or corrector.
- Title pages invoke ancient authorities to situate the edition within the classical tradition.



Between Restoration and Adaptation

- Many title pages emphasize **adaptation**: new demonstrations, diagrams, or reorganized content.
- **Accessibility**: Efforts to make content clearer, briefer, and easier to understand through simplified and expanded content and new notations.
- **Utility**: Increasing attention to practical application, especially in later editions, like surveying and navigation. Some works frame the Elements as part of professional training, especially for military personnel.



Between Restoration and Adaptation

- Goals such as accessibility and utility are interconnected, but not reducible to each other.
- The adaptations respond to emerging pedagogical needs while also reflecting contemporary scholarly currents, such as the growing acceptance of symbolic algebra, often associated with greater clarity.
- The adaptations did not replace the humanist concern with fidelity to Greek sources. The different intellectual tendencies coexisted, sometimes within the same edition.

Conclusions: Plurality and Variety

- The title pages present the Elements as a canonical text in motion, preserved as an ancient authority yet continually redeployed and reshaped.
- They reveal strong local roots set within active pan-European networks.
- They show a treatise embedded in universities and colleges, but also in civic offices, military settings and professional spheres, with the Jesuit Order serving as a prominent pan-European presence.
- They display a plurality of coexisting intellectual priorities, including restoration, clarity, accessibility, utility and innovation.

→ The title pages challenge neat stories of a unified Euclidean tradition and uncover a lively, varied landscape of mathematical communities and practices.

Methodological Takeaways

- Title pages are a mediating space where multiple intellectual, commercial, and pedagogical priorities intersect that offers a panoramic view.
- Paratextual analysis, still relatively uncommon in the history of mathematics, can serve as a tool for reconstructing both the intellectual and the social dimensions of early modern mathematical cultures.
- Combined with digital humanities techniques, it allow large-scale comparative analysis across editions and contexts.



Thank You

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